



The Human Capital — Artificial Intelligence Symbiosis and Economic Growth

Orlando Gomes¹ 

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Abstract

This paper contributes to the recent literature on the economic growth implications of automation and artificial intelligence (AI), by devising and exploring the dynamics of an analytical model featuring the gradual substitution of labor by automated processes. In the proposed setting, a symbiotic relation between the expansion of AI and the accumulation of human capital stands out as the driver of long-term sustained growth. Two versions of the model are scrutinized. In the first version, the representative capitalist is an optimal planner, while workers (who are heterogeneous regarding their productivity levels) are hand-to-mouth consumers. In the second version, every agent formulates an intertemporal plan and may, simultaneously, be a worker and an investor. This difference in context is vital for identifying the potential beneficiaries of the implementation of labor-saving technologies.

Keywords Automation · Artificial intelligence · Human capital · Productivity · Endogenous growth · Intertemporal optimization

JEL Classification O41 · O33 · J24 · E24

1 Introduction

This study contributes to the recent body of literature on the role of automation and artificial intelligence (AI) as drivers of sustained economic growth. Specifically, an analytical model is conceived and explored with the objective of highlighting the emergence of a potential symbiotic relation between the accumulation of human capital and the development of labor-saving technologies. This symbiosis works as a propeller of endogenous growth.

✉ Orlando Gomes
omgomes@iscal.ipl.pt

¹ Lisbon Accounting and Business School – Polytechnic University of Lisbon (ISCAL-IPL) and CEFAGE (Univ. Évora - ISCAL) Research Center, Av. Miguel Bombarda, 20, 1069-035 Lisbon, Portugal

The devised framework builds upon the crucial premise that labor is heterogeneous. Workers are endowed with distinct productivity levels, and they locate over a productivity distribution, which can take a variety of different forms. In this setting, the notion of productivity is intertwined with the concepts of skill, task, and job. The higher the level of productivity of the worker, the better is the respective set of skills, and the more sophisticated are the tasks the worker can perform, or the jobs she/he will be able to engage in. Under a competitive markets' assumption, the wage of the worker has correspondence on the marginal productivity of labor, and therefore the worker's wage is also associated with the task or job she/he is capable of performing, given the corresponding skills and productivity.

In the proposed setting, capital accumulation also occupies center stage. Regarding capital ownership, two distinct assumptions, with significantly different implications, are taken into consideration. In a first scenario, there is a representative capitalist, who optimally plans the trajectories of investment and consumption. The ultimate goal of this agent is to maximize the intertemporal utility of consumption, but the optimal choice requires engaging on investment over time in order to 'keep the business running'. In this setting, workers are hand-to-mouth consumers, i.e., they consume, at every time period, the exact amount of wage they receive. The second scenario does not make an explicit distinction between workers and capital owners. In this case, every agent in the economy is an optimal planner, meaning that everyone simultaneously consumes and saves in order to achieve the desired goal, which is the maximization of intertemporal utility. As one will discuss throughout the analysis, the two scenarios have distinct implications regarding who benefits from the eventual gains of the adoption of labor-saving technologies.

A central piece of the model's structure relates to the criterion that justifies the substitution of human labor by machines and AI. Such criterion is straightforward: when resorting to human labor to perform a task, firms (or a representative firm) incur in a cost that consists in the competitive wage that they must pay to workers. If firms are capable of undertaking the same task, with an identical outcome, through automated processes or AI, but incurring in lower costs, then labor should be replaced by machines and algorithms. Wages vary with productivity and, thus, with the performed tasks. Costs of automation and AI are also different across activities, but these differences do not obey to the same logic as the differences in productivity of human labor. Costs of automation and AI are attached to our modelling apparatus through the logic underlying the job polarization argument.¹

The notion of job polarization enters the model not only because the high costs of automation on the poles of the task distribution are explicitly modelled, but also because a dynamic version of the framework is considered, whose main feature is the progressive expansion of the influence of AI, namely in the direction of tasks that require sophisticated human thinking. As AI becomes more efficient and dethrones a progressively larger number of human jobs, workers are relocated to the jobs that still resist automation. Part of these workers are unable to acquire new

¹ See Sect. 2 for a brief literature review and a discussion about the concept, causes, and implications of job polarization.

competencies and they fall back to low productivity jobs; the other part assimilates new skills and is promoted to highly qualified jobs. Consequently, the labor productivity distribution is split in two, with low and high skill jobs separated by an AI gap (which contains the activities in which AI is dominant).

In the framework under analysis, sustained growth emerges as the outcome of the symbiotic relation between AI development and the shift of a portion of the relocated labor force to highly qualified jobs. The process is straightforward: the penetration of AI leads to better qualified jobs that allow workers to develop additional AI solutions that, once implemented, push part of the workers further to the right along the productivity distribution. This creates a never-ending loop that translated into endogenous sustained growth. The long-term balanced growth path (BGP) is characterized by the persistence of a positive and constant output growth rate.

Do all the individuals in the economy benefit from the progressive replacement of humans by machines and algorithms? The answer is contingent on the assumed economic structure and organization. In an economy where every single agent is an optimal planner, who works, consumes, and saves (invests), the transition towards the robot and AI economy is favorable to all (everyone will access earnings that grow as much as the economy). In an economy with a strict separation between capitalists and workers, with the latter being hand-to-mouth consumers, the income of the capital owner and the wages of high-skilled workers potentially grow at the same rate as the economy's output, while the wage of the low-skilled workers remains constant.

The remainder of the manuscript is organized as follows. Section 2 briefly reviews the growth theory literature that addresses the consequences of automation and AI. Section 3 starts building the model, in this first stage by considering a simple economy with heterogeneous labor and no AI. Section 4 addresses automation costs and the replacement of workers by machines. In Sect. 5, an analytical mechanism is proposed in order to characterize the relocation of labor triggered by technological advancements. Section 6 assesses the impact of the introduction of automation and AI under different scenarios regarding the ownership of capital. Section 7 focuses on the eventual expansion of the AI frontier and what this signifies for the dynamics of labor and machine allocation. Section 8 characterizes the economy's BGP under each of the adopted assumptions regarding capital ownership and the behavior of workers. Section 9 concludes.

2 Literature on Automation, AI, and Growth: a Brief Tour

2.1 A New Wave of Innovation

Economic growth history is also the history of technological breakthroughs. Whenever human ingenuity could generate and implement ideas with the potential to create value, the economy progressed and standards of living improved. The most recent waves of innovation, in which we can include, in a first stage, the intensive process of automation of routine activities and, in a second stage, the

fast dissemination of AI, are particularly impactful, in such a way that led Mokyr et al. (2015) to inquire whether this time will be different.

A significantly different economic landscape, built upon the new technological wonders, is likely to be formed. However, one cannot be sure about which direction the foreseeable changes will lead us. The pessimistic view emphasizes the fast and widespread substitution of labor by automated processes and algorithms, a movement that most likely will place a downward pressure on wages, generate massive unemployment, intensify income inequality, and increase personal anxieties and social unrest (Sachs & Kotlikoff, 2012; Sachs et al., 2015; Benzell et al., 2018; Korinek & Stiglitz, 2019, 2021). Evidence suggesting that robots are stealing jobs and that, in many businesses, human labor is becoming obsolete, abounds in the literature (DeCanio, 2016; Pellegrino et al., 2017; Bessen, 2019; Acemoglu & Restrepo, 2020a). Rapid AI progress increases fears and anxieties, not only those pertaining to the fall in the labor share, but also regarding market concentration, income distribution, and data protection and privacy (Ernst et al., 2019).

On the bright side, many researchers argue that AI should be interpreted as a new general-purpose technology with the potential to positively and pervasively impact the economy (Trajtenberg, 2019; Varian, 2019). AI may not only boost productivity, but it can also be a vital instrument in the creation of new and better jobs (Stevenson, 2019).

The astonishing pace at which AI has been evolving brought to the lexicon of economists a new word, *singularity*. Nordhaus (2021) defines singularity as the virtual threshold beyond which machines accomplish superintelligence, thus rendering all human labor obsolete. Although this is an interesting notion to discuss on conceptual grounds, the effective occurrence of a singularity is not perceived as a credible possibility in the near future. One of the reasons why one should not jump to the conclusion that AI will trigger a singularity is that, despite its prodigious capabilities, it has a relatively confined role as a new production input. Agrawal et al. (2019a, 2019b) and Babina et al. (2024) clarify that AI is, essentially, a prediction technology (i.e., AI allows firms to learn faster and better from large data sets). However, to generate value, businesses need more than prediction; they also require decision, and decision is eminently a human task (that humans should preserve as such).

Therefore, more than perceiving AI as a threat to human labor, agents in the economy can interpret it as an opportunity to shift workers from prediction to decision tasks. Under the optimistic view, one should expect the advancement of automation and AI, on the one hand, and the creation of more and better jobs, on the other hand, to reinforce each other (Brynjolfsson et al., 2023), even if this correlation is observed only with a significant time lag, something that is common to most general-purpose technologies (Brynjolfsson et al., 2019).

A plethora of authors interpret automation and AI as the seeds of a new economic era, whose shape is not yet well defined. When discussing the macroeconomic prospects of AI, we stand at a crossroad: effective policies can direct us to the virtuous path of productivity growth, fair income distribution, and labor-enhancing innovation; poorly devised policies and regulation may push us towards a vicious cycle

of market power concentration, strong and unfair inequality, and jobless societies (Acemoglu & Restrepo, 2020b; Brynjolfsson & Unger, 2023; Grilli et al., 2024).²

2.2 Job Polarization

Automation and AI contribute to growth via the job polarization effect that was mentioned in the Introduction. This effect has been widely discussed in the literature, both theoretically and empirically. The concept of job polarization is attached to the evidence that middle-skill/middle-wage jobs are disappearing in favor of automated processes, an evidence that is pervasive across labor markets all over the world (Goos et al., 2014; Graetz & Feng, 2015; Jaimovich & Siu, 2020). Underlying job polarization is the fact that middle-skill/middle-wage jobs correspond to routine codifiable tasks, for which the costs of automation are relatively low. Jobs on the edges of the skill distribution are more resistant to the penetration of robots and computer codes, in the sense that they represent tasks where, for distinct reasons, human intervention remains essential (Autor & Dorn, 2013; Autor et al., 2017).

On the right-end of the productivity distribution, highly sophisticated creative activities predominate, whose costs of effective automation are significant. On the left-end, costs of automation are high as well but, in this case, for a different motive. At this end, one finds activities that need a human touch, as child caring, house cleaning, or hairdressing. These are tasks that do not require overly refined skills, and, therefore, are not associated with high earnings. However, they demand a form of knowledge that machines do not possess, namely what one can designate as tacit knowledge (e.g., under the form of common sense, intuition, or empathy).

Autor (2014, 2015) highlights the apparent paradox involving, on the one hand, the simple tasks humans easily perform but that are not predisposed to codification (located at the left extremity of the skills' distribution), and, on the other hand, the computationally demanding tasks (located at the middle of the skills' distribution), for which machines typically exhibit a much better performance than humans. This is known as the Polanyi's paradox. The job polarization evidence led Autor (2022) to imagine a new division of labor, where machines and algorithms replace humans in any conceivable task that can be narrowed down to a set of instructions or procedures. The issue is that such codifiable activities are expanding fast, due to the impressive capabilities of AI. The AI revolution signifies the ability of machines to develop their own tacit knowledge, and therefore to penetrate tasks one would imagine to be unassailable for automation. In a certain sense, the new potentialities of AI are reversing Polanyi's paradox: the ability of machines to learn from big data is generating a new kind of tacit knowledge, which is, in many circumstances, inscrutable for humans.

² See Lu and Zhou (2021) for a review of the literature on the macroeconomic implications of AI.

2.3 Models of Automation and Growth

Growth models in which automation occupies center stage became the norm in contemporaneous economic theory. The simplest conceivable way of modelling the substitution of human labor by robots and algorithms consists in devising an aggregate production function in which automation capital, P , joins physical capital, K , and labor, L , as production inputs (Steigum, 2011; Prettnner, 2019; Lankisch et al., 2019; Antony & Klarl, 2020). In this context, automation capital emerges, within a standard Cobb-Douglas production function, as a perfect substitute for labor. Heer and Irmen (2019) highlight that the two forms of capital are also perfect substitutes, not as inputs in production but as stores of value. This signifies that the investment rates associated with each type of capital are endogenously determined, and coevolve along the transitional growth path.

In the aforementioned models, automation encloses the potential for long-run sustained growth, even in the absence of conventional technological progress. Because the economy accumulates two types of capital, decreasing marginal returns are potentially overcome, leading to endogenous growth. The downside is that jobs are lost, and capital owners concentrate income, which, in the new setting, corresponds to returns on two different forms of capital. The concentration of income and the progressive elimination of labor may have disruptive effects and hamper growth, e.g., through the contraction of aggregate demand.³

One pertinent doubt attached to the above discussion is whether empirical evidence effectively supports the idea of a pervasive replacement of human labor by automated devices and procedures and, if so, what are the main drivers underlying this substitution. Abeliatsky and Prettnner (2023) and Sasaki (2023) highlight that massive substitution of workers by automation capital is, in fact, occurring, although this process is mainly circumscribed to developed countries, where fertility rates are low and labor supply is in decline. The theoretical argument is straightforward: labor scarcity generates an upward pressure on wages, what makes firms to have stronger incentives to replace people by automation capital.⁴

An alternative interpretation that, notwithstanding, also places the emphasis on the structure of the production function, is the one offered by Acemoglu and Restrepo (2018, 2019a, 2019b). In the Acemoglu-Restrepo framework, there exists a series of tasks that can be performed by labor or capital. Automation is the process through which tasks are transferred from labor to capital, what corresponds to a displacement effect.

The negative impact of the displacement effect over labor demand and over the labor share is compensated by a few countervailing forces, that the mentioned authors group into four categories: (i) extensive-margin productivity effect (as the

³ Not all formal analyses of growth and automation point to sustained growth. Assuming an overlapping generations setting, Gasteiger and Prettnner (2022) suggest that the robot economy may converge to a zero-growth steady state equilibrium.

⁴ A similar argument, approached under different theoretical frameworks, is also explored by Irmen (2021) and Acemoglu and Restrepo (2022).

costs of automated tasks fall, the economy expands and the demand for labor in non-automated tasks increases); (ii) intensive-margin productivity effect (automation does not increase productivity only by replacing labor; it also increases productivity in tasks that have already been automated. This is a deepening of automation effect); (iii) Task complementarity effect (increased automation imposes, in many cases, the need for more labor); (iv) Reinstatement effect (creation of new tasks for which humans have a comparative advantage).

The mentioned authors place particular emphasis on the reinstatement effect. Overlooking the other channels, one might say that the net impact of automation on labor demand (and also on productivity and welfare) depends on the relative strength of each of the two enunciated effects: displacement and reinstatement. In this setting, automation and the creation of new tasks are closely intertwined: new technologies and AI need new tasks to operate more efficiently and require new high-skilled jobs to train, explain, and sustain AI. Therefore, the displacement and reinstatement effects go hand-in-hand and tend to reinforce each other.

With the above-described framework in mind, Acemoglu (2024) conjectures that the exact shape of the future of work is not, in any way, pre-established or pre-determined. On the contrary, it is strongly dependent on how technologies are employed to modify the task content of production. At this level, policy matters. With adequate labor market policies, the reinstatement effect can overcome the displacement effect, thus guaranteeing that labor continues to be pivotal for the generation of income and for economic growth.

3 Heterogeneous Labor Productivity and Capital Ownership

Conceive an economy populated by a large number of heterogeneous workers. Heterogeneity is associated with labor productivity and, concomitantly, with the tasks or jobs that workers are capable of performing. The productivity distribution is represented by a continuous probability density function (PDF), $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. The argument of the function, $x \geq 0$, denotes workers' productivity. The density of workers endowed with productivity level x is given by $f(x)$. In this setting, the total amount of labor is normalized to 1, i.e., $L = \int_0^{+\infty} f(x)dx = 1$. Aggregate labor productivity writes as $A_L = \int_0^{+\infty} f(x)xdx$.

Whenever feasible, the analysis is undertaken for a general productivity distribution with the above-mentioned features. However, this is not always possible, and therefore, at some stages of the analysis, specific configurations are adopted. Namely, one considers a Weibull distribution with PDF $f(x;\lambda,k) = \lambda k(\lambda x)^{k-1} e^{-(\lambda x)^k}$; $\lambda > 0$ and $k > 0$ are, respectively, an inverse scale parameter and a shape parameter.

Illustrations will be circumscribed to two cases, which take, alternatively, $k = 1$ and $k = 2$. Function $f(x;\lambda,1)$ corresponds to the special case of the exponential distribution. This distribution represents a setting in which the number of workers with capacity to perform a given job progressively declines with the complexity of the job, i.e., $f' < 0$ for the entire range of productivity levels. In this case, the inverse scale parameter indicates the extent of concentration or dispersion of workers across

jobs (the larger the value of λ , the stronger is the concentration of the workforce in low-skilled tasks). The second case corresponds to a PDF $f(x; \lambda, 2)$ displaying a hump-shaped configuration, which reflects a concentration of workers in middle-skill jobs. These high-density middle-skill jobs locate as much to the right of the productivity distribution as the lower is the value of λ .

For matters of exposition, hereafter the two illustrative examples are identified as *Exp* and *Wei*, respectively. For each of the two distributions, aggregate labor productivity can be explicitly presented: $A_L = \lambda^{-1} (\text{Exp})$; $A_L = \frac{\sqrt{\pi}}{2} \lambda^{-1} (\text{Wei})$. For the same inverse scale parameter value, aggregate productivity is larger in the *Exp* case.

Concerning the decisions and behavior of economic agents, two different settings will be pondered. In the first setting, a complete separation between workers and capital owners exists. To make this separation operational, one assumes that workers behave as hand-to-mouth consumers, i.e., they do not save and, therefore, they never turn into capitalists.⁵ In the second scenario, all workers are optimal planners and, therefore, they are simultaneously consumers and investors.

Let us start by approaching the first scenario, where every worker spends her/his current wage in current consumption, $c(x) = w(x)$, with $c(x)$, $w(x)$ representing, respectively, the consumption and the wage of a worker endowed with productivity level x . Further below, an expression for $w(x)$ is derived. Unlike workers, capital owners in this economy are homogeneous. This signifies that the analysis pertaining to this class of agents can be reduced to the discussion of the behavior of a representative agent. The capital owner is an intertemporal optimizer, who chooses the paths of consumption and investment that maximize utility. Specifically, the capital owner maximizes the present value of consumption utility, given an infinite horizon,

$$V_0 = \sum_{t=0}^{+\infty} \left(\frac{1}{1+\rho} \right)^t u(C_t) \quad (1)$$

In expression (1), $\rho \geq 0$ stands for the rate of time preference, and $u(C_t)$ is a constant intertemporal elasticity of substitution (CIES) utility function, with consumption as the single argument. The maximization of V_0 in (1) is subject to a capital accumulation constraint, which takes a trivial form,

$$K_{t+1} = (1 + r_t - \delta)K_t - C_t, \quad K_0 \text{ given} \quad (2)$$

In Eq. (2), r_t represents the gross rate of return on capital and $\delta \in (0, 1)$ is the capital depreciation rate. The stock of physical capital at date t is represented by $K_t \geq 0$. Assuming competitive markets for production inputs, the wage rate and the capital rate of return correspond, respectively, to the marginal product of labor and to the marginal product of capital. To compute these marginal products, one needs to characterize the technology of production. For this purpose, define the following

⁵ There is plenty of empirical evidence suggesting hand-to-mouth behavior among a large percentage of workers, even those with high wages. The wealthy hand-to-mouth argument elaborated by Kaplan et al. (2014) points in this direction.

Cobb-Douglas production function with labor-augmenting or Harrod-neutral technological progress,

$$Y = K^\alpha A_L^{1-\alpha} \quad (3)$$

Parameter $\alpha \in (0, 1)$ stands for the output elasticity of capital. Recalling the definition of A_L , earlier in this section, one may single out individual worker $\ell(x)$ as someone who holds productivity level x and, thus, belongs to the group $f(x)$ of workers that share the same productivity.

The wage of the individual worker, $\ell(x)$, and the rate of return on the capital invested by the capital owner are, respectively,

$$w(x) = \frac{\partial Y}{\partial \ell(x)} = (1 - \alpha) \left(\frac{K}{A_L} \right)^\alpha x \quad (4)$$

$$r = \frac{\partial Y}{\partial K} = \alpha \left(\frac{K}{A_L} \right)^{-(1-\alpha)} \quad (5)$$

Observe that the wage of the worker is proportional to the corresponding productivity. The aggregate wage paid to workers in the economy amounts to: $W = \int_0^{+\infty} w(x)f(x)dx = (1 - \alpha)Y$. Besides wages, the remaining part of aggregate income is the capitalist's return, i.e., $rK = \alpha Y$.

The solution of the capitalist's problem is trivial. By applying conventional optimal control techniques, and taking a logarithmic utility function, one obtains the following equation of motion for consumption,⁶

$$C_{t+1} = \frac{1}{1 + \rho} \left[1 - \delta + \alpha^2 \left(\frac{K_{t+1}}{A_L} \right)^{-(1-\alpha)} \right] C_t \quad (6)$$

For the pair of difference Eqs. (6) and (2), it is straightforward to compute the steady state solution concerning the income variable, which is,

$$Y^* = \left(\frac{\alpha^2}{\rho + \delta} \right)^{\alpha/(1-\alpha)} A_L \quad (7)$$

Result (7) suggests that the rate at which income grows in the steady state coincides with the growth rate of labor productivity. Thus far, aggregate productivity has been kept constant and, therefore, income (under the form of wages for workers and capital returns for the capital owner) is constant as well.

In the second setup, in which optimal planning is pervasive, every worker is endowed with an initial wealth level, $k_0(x) \geq 0$, that grows over time as commanded by the following constraint:

⁶ See Appendix 1 for the derivation of Eq. (6).

$$k_{t+1}(x) = w_t(x) + (1 + r_t - \delta)k_t(x) - c_t(x), \quad k_0(x) \text{ given.} \quad (8)$$

The configuration of the aggregate production function continues to determine the wage and the rate of return on capital, as displayed in (4) and (5). Hence, Eq. (8) can be rewritten as

$$k_{t+1}(x) = \left[(1 - \alpha) \frac{x}{A_L} + \alpha \frac{k_t(x)}{K_t} \right] K_t^\alpha A_L^{1-\alpha} + (1 - \delta)k_t(x) - c_t(x) \quad (9)$$

In (9), aggregate physical capital is defined as the sum of the capital endowment of every agent, i.e., $K_t = \int_0^{+\infty} f(x)k_t(x)dx$.

Let $V_0(x) = \sum_{t=0}^{+\infty} \left(\frac{1}{1+\rho} \right)^t u[c_t(x)]$ be the intertemporal utility function of a worker endowed with productivity x . The individual worker solves the optimality problem that consists in the maximization of $V_0(x)$ subject to constraint (9). The solution is the dynamic equation for consumption,⁷

$$c_{t+1}(x) = \frac{1}{1+\rho} \left((1 - \delta) + \alpha \left\{ 1 + (1 - \alpha) \left[\frac{x}{A_L} - \frac{k_{t+1}(x)}{K_{t+1}} \right] \right\} \left(\frac{K_{t+1}}{A_L} \right)^{-(1-\alpha)} \right) c_t(x) \quad (10)$$

The evaluation of Eq. (10) in the steady state equilibrium yields

$$\frac{k^*(x)}{K^*} = \frac{1}{1-\alpha} + \frac{x}{A_L} - \frac{\rho + \delta}{\alpha(1-\alpha)} \left(\frac{K^*}{A_L} \right)^{1-\alpha} \quad (11)$$

Equation (11) indicates that the relative stock of capital of an agent with productivity x is, in the steady state, as much higher as the higher is the corresponding productivity level. Aggregating over all agents, one computes the economy's steady state stock of capital and, given the production function, also the steady state level of income,

$$K^* = \left(\frac{\alpha}{\rho + \delta} \right)^{1/(1-\alpha)} A_L \quad (12)$$

$$Y^* = \left(\frac{\alpha}{\rho + \delta} \right)^{\alpha/(1-\alpha)} A_L \quad (13)$$

One can compare income solutions in (7) and (13). This latter value is larger than the former. Such outcome is the direct consequence of assuming that in this second scenario every agent behaves optimally, while in the first setting only the capital owner adopts such behavior (the hand-to-mouth behavior of workers is not, by definition, optimal). This comparison is meaningful because it highlights how damaging

⁷ We refrain from presenting the derivation of the worker's consumption equation. The procedure to obtain it is similar to the one explained in Appendix 1 for aggregate consumption.

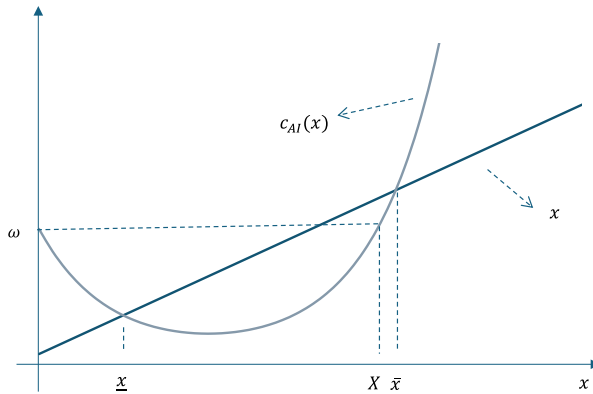


Fig. 1 The replacement rule: marginal product vs automation cost

it is for long-term income distribution the inability or the unwillingness of the agents in adopting optimal behavior. Such harmful effect is exacerbated with the introduction of labor-saving technologies, as characterized in the following sections.

4 The Codification of Work: Human Touch vs Human Creativity

Regardless of who the capital owners are, consider that production takes place in a representative firm under the supervision of a representative manager, whose goal is to maximize the firm's value. To attain such desideratum, the manager's decision on whether to maintain labor in the development of specific tasks or, alternatively, replace it by automated processes, obeys to a simple criterion: for a given task or job, if the costs of development and implementation of the labor-saving technology are lower than the marginal product generated by such task, then automation and AI solutions should substitute for labor (in this case, the firm will no longer incur in the cost of paying the wage to the worker, and therefore it benefits from the substitution). Representing the costs of research and implementation of automation and AI by $c_{AI}(x)$, the rule for the replacement of humans by machines is $c_{AI}(x) < x$.

Following the job polarization argument, discussed in Sect. 2, we consider that there are two types of costs associated with automation: (i) human touch costs, $c_{AI,ht}(x)$, which are larger for relatively low values of x , and (ii) human creativity costs, $c_{AI,hc}(x)$, which are larger for relatively high values of x . Cost functions are formulated as follows,

$$c_{AI,ht}(x) = \omega e^{-\omega x}, \quad \omega > 0 \quad (14)$$

$$c_{AI,hc}(x) = \omega e^{-\omega(X-x)}, \quad \omega > 0, X > 0 \quad (15)$$

Total costs pertaining to automation and AI amount to $c_{AI}(x) = c_{AI,ht}(x) + c_{AI,hc}(x)$. The replacement threshold is the solution of the following equation,

$$\omega[e^{-\omega x} + e^{-\omega(X-x)}] = x \quad (16)$$

Figure 1 represents both terms of Eq. (16), for the conceivable range of jobs or tasks. The r.h.s. of the equation, which corresponds to the marginal product, is a linear increasing function of x ; the l.h.s., which corresponds to the cost function, is a U-shaped curve that intersects the productivity line at two points: $\underline{x}$ and $\bar{x}$. In the interval between these two points lie the jobs that will no longer be performed by human labor, because the firm finds it advantageous to replace people by automated processes, which are, in this case, less expensive. Note the job polarization feature: less expensive automation occurs for activities somewhere in intermediate positions of the productivity or job sophistication scale. At the two ends of the productivity distribution, jobs continue to be performed by human labor, given the high costs of automation associated with such tasks.

Jobs lost to automation and AI amount to $\int_{\underline{x}}^{\bar{x}} f(x)dx$. This share of displaced jobs differs with the configuration of the productivity distribution. Recovering the two previously presented distributions, one computes the following terms,

$$\int_{\underline{x}}^{\bar{x}} f(x)dx = e^{-\lambda \underline{x}} - e^{-\lambda \bar{x}} \quad (Exp) \quad (17)$$

$$\int_{\underline{x}}^{\bar{x}} f(x)dx = e^{-(\lambda \underline{x})^2} - e^{-(\lambda \bar{x})^2} \quad (Wei) \quad (18)$$

To shares of workers (17) and (18) correspond productivity levels equal to:

$$\int_{\underline{x}}^{\bar{x}} f(x)x dx = \frac{(1 + \lambda \underline{x})e^{-\lambda \underline{x}} - (1 + \lambda \bar{x})e^{-\lambda \bar{x}}}{\lambda} \quad (Exp) \quad (19)$$

$$\int_{\underline{x}}^{\bar{x}} f(x)x dx = \frac{\sqrt{\pi}}{2} \frac{[\text{erf}(\lambda \bar{x}) - \text{erf}(\lambda \underline{x})]}{\lambda} + \underline{x}e^{-(\lambda \underline{x})^2} - \bar{x}e^{-(\lambda \bar{x})^2}, \quad (20)$$

with $\text{erf}(\lambda x) = \frac{2}{\sqrt{\pi}} \int_0^{\lambda x} e^{-z^2} dz$ the error function (Wei)

The substitution of workers for automated processes will imply that the firm preserves productivity level $\int_{\underline{x}}^{\bar{x}} f(x)x dx$ in the interval $(\underline{x}, \bar{x})$, but now it incurs on automation and AI costs that lower the potential outcome. The net productivity of AI is, then,

$$A_{AI} = \int_{\underline{x}}^{\bar{x}} f(x)[x - c_{AI}(x)] dx \quad (21)$$

If the labor that becomes obsolete in the original tasks it was assigned to is not relocated to other jobs, there is a loss of aggregate income. The lost wages exceed the

gain of the capital owner, who now extracts income not only from physical capital but also from the machines and algorithms that replace manpower.

In the new AI economy, the production function becomes

$$Y = K^\alpha \left(A_{\underline{L}} + A_{AI} + A_{\bar{L}} \right)^{1-\alpha} \quad (22)$$

with $A_{\underline{L}} = \int_0^{\underline{x}} f(x)xdx$ and $A_{\bar{L}} = \int_{\bar{x}}^{+\infty} f(x)xdx$ representing the aggregate productivity of workers allocated to jobs that persist, respectively, at the left-end and at the right-end of the productivity distribution. The total wages paid to workers correspond, in the current scenario, to

$$\begin{aligned} W &= \int_0^{\underline{x}} w(x)f(x)dx + \int_{\bar{x}}^{+\infty} w(x)f(x)dx \\ &= (1 - \alpha) \frac{A_{\underline{L}} + A_{\bar{L}}}{A_{\underline{L}} + A_{AI} + A_{\bar{L}}} Y \end{aligned} \quad (23)$$

The aggregate wage has fallen, relative to the no AI setup, in the extent of the displacement of labor to automated activities [recall that prior to automation, $W = (1 - \alpha)Y$]. The return on capital (both physical and automation) is,

$$rK + \frac{\partial Y}{\partial A_{AI}} A_{AI} = \frac{\alpha \left(A_{\underline{L}} + A_{\bar{L}} \right) + A_{AI}}{A_{\underline{L}} + A_{AI} + A_{\bar{L}}} Y \quad (24)$$

Comparing (24) with the capital return without automation, i.e., αY , one confirms the gain for capitalists that emerges from labor displacement. As stated, the overall outcome is now worse than in the original scenario, because total income is lower under (22) than under the original production function; this is true because $A_{\underline{L}} + A_{AI} + A_{\bar{L}} < A_{\underline{L}}$.

5 Labor Reinstatement: Can Humans Strike Back?

As suggested by Acemoglu's reinstatement argument, jobs lost to AI - those in the productivity interval $(\underline{x}, \bar{x})$ - are likely to be relocated elsewhere. The most reasonable assumption is to consider that the labor force originally in such productivity range splits into two groups: those that are unable to assimilate new competencies and, therefore, are pushed back to tasks requiring a less sophisticated set of skills (i.e., tasks for which x is low); and those who take advantage of the need to recon-vert and are capable of evolving to high-skilled jobs (i.e., jobs for which x is high). Hence, labor is partially reinstated into the left part of the original distribution and partially reinstated into the right segment of the distribution, leading to a skill gap that did not exist prior to automation.

In the subsequent analysis, labor relocation is modelled under two different assumptions, namely:

- (i) Workers are reassigned to new tasks such that each of the two remaining segments of the productivity distribution move up proportionally (designate this as the *MR* relocation process);
- (ii) Workers are reassigned to new tasks such that each of the two remaining segments of the productivity distribution preserve the characteristics of the original distribution (call this the *OR* relocation process).

Possible reinstatement processes are depicted in Fig. 2, for the two aforementioned scenarios, and taking into consideration the two prototypical labor productivity distributions adopted earlier. The figure has four quadrants. In the top panels, the proportional reassignment for the *Exp* and the *Wei* distributions is displayed; in this case, dismissed workers are proportionally relocated to other tasks, at both ends of the productivity distribution. In the bottom panels, the distribution-preserving reinstatement is represented. The left panels are drawn for the *Exp* and the right panels for the *Wei* distributions, respectively.

The growth analysis in subsequent sections will uncover that the different assumptions regarding labor relocation are not decisive in determining either the pace of long-term economic growth or the distribution of gains in the AI economy. These are predominantly shaped by the alternative assumptions concerning capital ownership and consumption-savings choices.

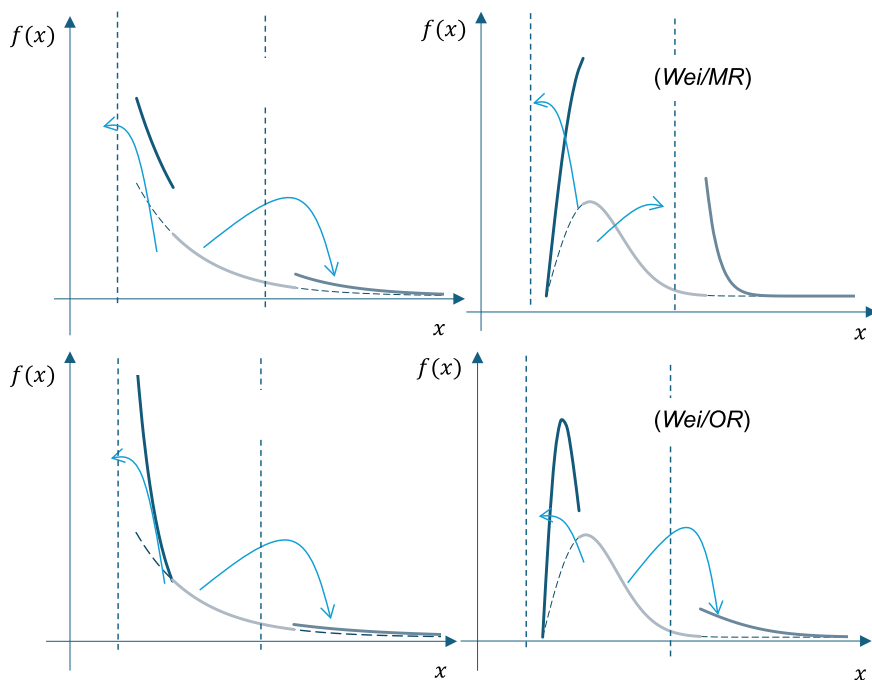


Fig. 2 Reinstatement of labor

Start by assuming that labor relocation obeys the proportionality rule. Because the preserved segments of the labor productivity distribution shift up, in the same proportion of the original distribution of workers per task, one can define a new pair of parameters, $\hat{\zeta}$ and $\tilde{\zeta}$, such that:

$$\int_0^{\underline{x}} (1 + \hat{\zeta}) f(x) dx + \int_{\underline{x}}^{+\infty} (1 + \tilde{\zeta}) f(x) dx = 1 \quad (25)$$

The first integral in (25) respects to the amount of labor that is now associated with low-skilled tasks, while the second integral corresponds to the labor attached to high-skilled activities. Imagining that all workers replaced by automated processes get a new job, the total workforce engaged in production remains equal to 1.

Let $\eta \in (0, 1)$ be the share of labor lost to low-skilled jobs, following the automation and AI shock; $1 - \eta$ is the share of labor lost to high-skilled jobs. Given η , the following relations hold,

$$\begin{aligned} \int_0^{\underline{x}} (1 + \hat{\zeta}) f(x) dx &= \int_0^{\underline{x}} f(x) dx + \eta \int_{\underline{x}}^{\bar{x}} f(x) dx \\ \Rightarrow \hat{\zeta} &= \eta \frac{\int_{\underline{x}}^{\bar{x}} f(x) dx}{\int_0^{\underline{x}} f(x) dx} \end{aligned} \quad (26)$$

$$\begin{aligned} \int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta}) f(x) dx &= (1 - \eta) \int_{\underline{x}}^{\bar{x}} f(x) dx + \int_{\bar{x}}^{+\infty} f(x) dx \\ \Rightarrow \tilde{\zeta} &= (1 - \eta) \frac{\int_{\underline{x}}^{\bar{x}} f(x) dx}{\int_{\bar{x}}^{+\infty} f(x) dx} \end{aligned} \quad (27)$$

The total productivity of the low-skilled (those to the left of the automated tasks) and of the high-skilled (those to the right of the automated tasks) is now, respectively, $A_{\underline{L}} = \int_0^{\underline{x}} (1 + \hat{\zeta}) f(x) x dx$ and $A_{\bar{L}} = \int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta}) f(x) x dx$. Total productivity of labor and automated processes is $A_{\underline{L}} + A_{AI} + A_{\bar{L}}$, with A_{AI} given by (21). This value should be compared with the original productivity of labor, A_L in order to understand whether the economy as a whole gains or loses with the reorganization of economic activity triggered by the automation of processes. This comparison is unfeasible to undertake in general terms; a numerical example is explored below, after the characterization of the *OR* case.

To explore scenario *OR*, recover the general Weibull PDF $f(x; \lambda, k)$. If one wants to consider a process of relocation of labor that preserves the configuration of the original distribution for both segments of the post-AI distribution, one must consider two new inverse scale parameters, $\hat{\lambda}$ and $\tilde{\lambda}$, such that

$$\int_0^{\underline{x}} f(x; \hat{\lambda}, k) dx + \int_{\bar{x}}^{+\infty} f(x; \tilde{\lambda}, k) dx = 1 \quad (28)$$

Assuming the same share η as for case *MR*, the following identities hold,

$$\int_0^{\underline{x}} f(x; \hat{\lambda}, k) dx = \int_0^{\underline{x}} f(x; \lambda, k) dx + \eta \int_{\underline{x}}^{\bar{x}} f(x; \lambda, k) dx \quad (29)$$

$$\int_{\bar{x}}^{+\infty} f(x; \tilde{\lambda}, k) dx = (1 - \eta) \int_{\underline{x}}^{\bar{x}} f(x; \lambda, k) dx + \int_{\bar{x}}^{+\infty} f(x; \lambda, k) dx \quad (30)$$

Under both assumed distributions, *Exp* and *Wei*, Eq. (28) imposes constraint $\hat{\lambda}_{\underline{x}} = \tilde{\lambda}_{\bar{x}}$. Equations (29) and (30) allow for the presentation of explicit solutions for the inverse scale parameters regarding each of the two distributions. The corresponding formulas are displayed below for $\hat{\lambda}$ (to compute $\tilde{\lambda}$ just consider the above equality involving the two parameters and the boundaries of the AI productivity interval),

$$\hat{\lambda} = -\frac{1}{\underline{x}} \ln \left[(1 - \eta) e^{-\lambda \underline{x}} + \eta e^{-\lambda \bar{x}} \right] \quad (Exp) \quad (31)$$

$$\hat{\lambda} = \frac{1}{\underline{x}} \sqrt{-\ln \left[(1 - \eta) e^{-(\lambda \underline{x})^2} + \eta e^{-(\lambda \bar{x})^2} \right]} \quad (Wei) \quad (32)$$

Productivity levels correspond, in the *OR* setting, to $A_{\underline{L}} = \int_0^{\underline{x}} f(x; \hat{\lambda}, k) x dx$ and $A_{\bar{L}} = \int_{\bar{x}}^{+\infty} f(x; \tilde{\lambda}, k) x dx$; A_{AI} is still given by (21).

The characterized productivities can be explicitly quantified if one assigns concrete values to the parameters in the model. In the illustration that follows, let $\omega = 2$, $X = 2.5$, $\lambda = 1$. A first straightforward outcome emerging from the example respects the values of the lower and upper bounds of x ; recovering job replacement rule (16), one calculates $\underline{x} = 0.6012$ and $\bar{x} = 5.5064$. Given these values and the definitions of the productivity levels, Table 1 displays the results obtained for $A_{\underline{L}}$, A_{AI} , and $A_{\bar{L}}$, and for the sum of these three components. Results are displayed for differently shaped productivity distributions, for different rules of labor reinstatement, and for a sample of possible values of parameter η .

Results in the last columns of Table 1, for total productivity of labor plus automation, should be compared with labor productivity in the absence of automation. Recall that, in the pre-AI scenario, $A_L = \lambda^{-1} = 1$ (*Exp*) or $A_L = \frac{\sqrt{\pi}}{2} \lambda^{-1} = 0.8862$ (*Wei*). The observation of the figures in the table indicates that, in general, $A_{\underline{L}} + A_{AI} + A_{\bar{L}} > A_L$. The opposite result occurs in very specific circumstances (namely, for the *Wei* distribution when values of η are very close to 1). Results are graphically presented in Fig. 3. Two panels are displayed, one for each type of distribution. In both cases, A_L is compared with

Table 1 Productivity levels of labor and machine-AI, after labor reinstatement

Exp / MR	A_L	A_{AI}	A_L	$A_L + A_{AI} + A_L$
$\eta = 0$	0.1294	0.6704	1.9510	2.7508
$\eta = 0.25$	0.1618	0.6704	1.5284	2.3606
$\eta = 0.5$	0.1943	0.6704	1.1057	1.9704
$\eta = 0.75$	0.2267	0.6704	0.6831	1.5802
$\eta = 1$	0.2592	0.6704	0.2605	1.1901
Exp / OR	A_L	A_{AI}	A_L	$A_L + A_{AI} + A_L$
$\eta = 0$	0.1294	0.6704	3.6863	4.4861
$\eta = 0.25$	0.1547	0.6704	2.3865	3.2116
$\eta = 0.5$	0.1746	0.6704	1.4750	2.3207
$\eta = 0.75$	0.1854	0.6704	0.7914	1.6473
$\eta = 1$	0.1747	0.6704	0.2605	1.1055
Wei / MR	A_L	A_{AI}	A_L	$A_L + A_{AI} + A_L$
$\eta = 0$	0.1282	0.3004	1.9109	2.3394
$\eta = 0.25$	0.1957	0.3004	1.4338	1.9299
$\eta = 0.5$	0.2633	0.3004	0.9568	1.5205
$\eta = 0.75$	0.3309	0.3004	0.4797	1.1110
$\eta = 1$	0.3985	0.3004	0.0027	0.7016
Wei / OR	A_L	A_{AI}	A_L	$A_L + A_{AI} + A_L$
$\eta = 0$	0.1294	0.3004	3.6863	4.1161
$\eta = 0.25$	0.1508	0.3004	2.5706	3.0217
$\eta = 0.5$	0.1669	0.3004	1.8306	2.2979
$\eta = 0.75$	0.1797	0.3004	1.2209	1.7010
$\eta = 1$	0.1747	0.3004	0.2605	0.7355

total productivity in the automation scenario, and considering each of the reinstatement rules *MR* and *OR*.

6 Income Distribution in the Algorithmic Economy

The displacement and reinstatement effects, as characterized in the previous two sections, can be analyzed in light of each of the two formulated assumptions about capital ownership. Let us start by discussing the implications for the model with hand-to-mouth workers and the representative capitalist. In this case, aggregate wages are given by (23) and, therefore, this expression also indicates the aggregate level of consumption of the labor class. Returns on both forms of capital, physical and AI, are expressed in Eq. (24); these correspond to the returns of the capitalist who optimally chooses how

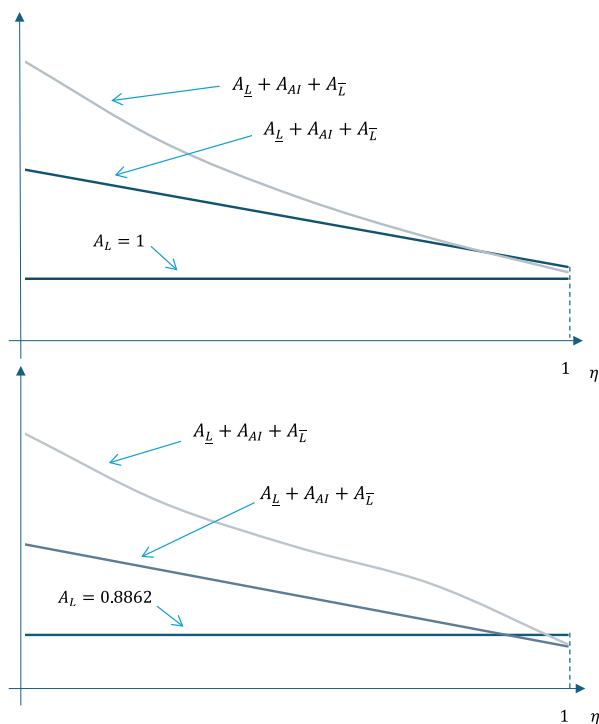


Fig. 3 Labor and machine-AI productivity

much to consume and how much to reinvest in production. The capital accumulation equation is now

$$K_{t+1} = (1 + r_t - \delta)K_t + \frac{\partial Y_t}{\partial A_{AI}} A_{AI} - C_t \Leftrightarrow$$

$$K_{t+1} = \frac{\alpha \left(A_L + A_L \right) + A_{AI}}{\left(A_L + A_{AI} + A_L \right)^\alpha} K_t^\alpha + (1 - \delta)K_t - C_t \quad (33)$$

The difference between Eqs. (2) and (33) is that returns from AI are, in the current setting, part of the capitalist's income, what leads to a modification of the productivity term in the constraint. This has direct implications on the optimal path of consumption, and also on steady state results, namely concerning aggregate income,

$$Y^* = \left[\frac{\alpha \left(A_L + A_L \right) + A_{AI}}{\rho + \delta} \right]^{\alpha/(1-\alpha)} \left(A_L + A_{AI} + A_L \right) \quad (34)$$

Because $\frac{\alpha(A_L + A_L^-) + A_{AI}}{A_L + A_{AI} + A_L^-} > \alpha$ and as long as $A_L + A_{AI} + A_L^- > A_L$, it is possible to guarantee that the new income level (34) is larger than steady state income (7).⁸

Next, resume the optimal planners' version. In this version of the model, every agent is simultaneously a consumer and an investor, as explained in Sect. 3. Because now agents also benefit from returns on automation and AI, an additional assumption is required. The assumption is that the earnings from automation and AI are proportional to the capital share of the agent, such that capital accumulation constraint (8) is modified in the following way,

$$k_{t+1}(x) = w_t(x) + (1 + r_t - \delta)k_t(x) + r_{AI,t}A_{AI} \frac{k_t(x)}{K_t} - c_t(x) \quad (35)$$

Given production function (22), the relevant returns on inputs are:

$$w_t(x) = (1 - \alpha)K_t^\alpha (A_L + A_{AI} + A_L^-)^{-\alpha} x \quad (36)$$

$$r_t = \alpha K_t^{-(1-\alpha)} (A_L + A_{AI} + A_L^-)^{1-\alpha} \quad (37)$$

$$r_{AI,t} = (1 - \alpha)K_t^\alpha (A_L + A_{AI} + A_L^-)^{-\alpha} \quad (38)$$

Replace (36), (37), and (38) in (35), to obtain the capital accumulation constraint that applies to the current case,

$$k_{t+1}(x) = \left[\frac{(1 - \alpha)x}{A_L + A_{AI} + A_L^-} + \frac{\alpha(A_L + A_L^-) + A_{AI}}{A_L + A_{AI} + A_L^-} \frac{k_t(x)}{K_t} \right] K_t^\alpha (A_L + A_{AI} + A_L^-)^{1-\alpha} + (1 - \delta)k_t(x) - c_t(x) \quad (39)$$

Solving the intertemporal problem for an agent endowed with productivity level x , i.e., the problem that consists of the maximization of the objective function $V_0(x)$ subject to constraint (39), the following dynamic equation for consumption is derived,⁹

⁸ Appendix 2 explores further the obtained results, resorting to an illustrative numerical example.

⁹ We refrain from presenting the steps required to obtain Eq. (40), since these are similar to the ones described in Appendix 1 when solving the optimal control problem for the representative capitalist.

$$c_{t+1}(x) = \frac{1}{1+\rho} \left(\left\{ \alpha(1-\alpha)x + \left[1 - (1-\alpha) \frac{k_{t+1}(x)}{K_{t+1}} \right] \left[\alpha(A_L + A_L^-) + A_{AI} \right] \right\} \times \frac{K_t^{-(1-\alpha)} (A_L + A_{AI} + A_L^-)^{-\alpha}}{K_t^{-(1-\alpha)} (A_L + A_{AI} + A_L^-)^{-\alpha}} \right) c_t(x) \quad (40)$$

The evaluation of (40) in the steady state allows for the derivation of the ratio between individual capital endowment and aggregate physical capital, which can be compared with (11),

$$\frac{k^*(x)}{K^*} = \frac{1}{1-\alpha} + \frac{\alpha x}{\alpha(A_L + A_L^-) + A_{AI}} - \frac{(\rho + \delta)(A_L + A_{AI} + A_L^-)^\alpha}{(1-\alpha)[\alpha(A_L + A_L^-) + A_{AI}]} (K^*)^{1-\alpha} \quad (41)$$

As one would expect, the higher the level of productivity of the agent, the larger is the ratio of the individual capital endowment relative to the aggregate level.

To proceed with the analysis, one needs to focus on the macro perspective, namely on the computation of total income and of income per class of agents. Recall that, in this version of the model, every individual is simultaneously a worker and a capital owner; however, as automation and AI penetrate the economy, they are separated into the two groups: the low-skilled and the high-skilled. The income of each of these groups can be denoted, respectively, by $\underline{Y}_t$ and $\overline{Y}_t$, and they correspond, by definition, to:

$$\underline{Y}_t = \int_0^{\underline{x}} \left[w_t(x) + (1+r_t-\delta)k_t(x) + r_{AI,t}A_{AI} \frac{k_t(x)}{K_t} \right] dx \quad (42)$$

$$\overline{Y}_t = \int_{\overline{x}}^{+\infty} \left[w_t(x) + (1+r_t-\delta)k_t(x) + r_{AI,t}A_{AI} \frac{k_t(x)}{K_t} \right] dx \quad (43)$$

The computation of the integrals yields,

$$\underline{Y}_t = \left\{ \frac{(1-\alpha)A_L}{A_L + A_{AI} + A_L^-} + \frac{\alpha(A_L + A_L^-) + A_{AI}}{A_L + A_{AI} + A_L^-} \frac{K_t(\underline{x})}{K_t} \right\} Y_t \quad (44)$$

$$\overline{Y}_t = \left\{ \frac{(1-\alpha)A_L^-}{A_L + A_{AI} + A_L^-} + \frac{\alpha(A_L + A_L^-) + A_{AI}}{A_L + A_{AI} + A_L^-} \frac{K_t(\overline{x})}{K_t} \right\} Y_t \quad (45)$$

In expressions (44) and (45), terms $K_t(\underline{x})$ and $K_t(\overline{x})$ correspond to the capital endowments of the classes of low-skilled workers and high-skilled workers, respectively.

These terms are defined as $K_t(\underline{x}) = \int_0^{\underline{x}} (1 + \hat{\zeta}) f(x) k_t(x) dx$ and

$K_t(\bar{x}) = \int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta})f(x)k_t(x)dx$, for the *MR* case, and as $K_t(\underline{x}) = \int_0^{\underline{x}} f(x;\hat{\lambda}, k)k_t(x)dx$ and $K_t(\bar{x}) = \int_{\bar{x}}^{+\infty} f(x;\tilde{\lambda}, k)k_t(x)dx$, for the *OR* case. The total stock of capital is, obviously, $K_t = K_t(\underline{x}) + K_t(\bar{x})$ and, from (44) and (45), it is also evident that $Y_t = \underline{Y}_t + \bar{Y}_t$.

Return to the steady state expression (41). Aggregating over the stocks of capital of each of the two groups of agents, the derived results are:

$$\frac{K^*(\underline{x})}{K^*} = \frac{\alpha A_{\underline{L}} + \left[\int_0^{\underline{x}} (1 + \hat{\zeta})f(x)dx \right] A_{AI}}{\alpha (A_{\underline{L}} + A_{\bar{L}}) + A_{AI}} \quad (MR) \quad (46)$$

$$\frac{K^*(\underline{x})}{K^*} = \frac{\alpha A_{\underline{L}} + \left[\int_0^{\underline{x}} f(x;\hat{\lambda}, k)dx \right] A_{AI}}{\alpha (A_{\underline{L}} + A_{\bar{L}}) + A_{AI}} \quad (OR) \quad (47)$$

$$\frac{K^*(\bar{x})}{K^*} = \frac{\left[\int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta})f(x)dx \right] A_{AI} + \alpha A_{\bar{L}}}{\alpha (A_{\underline{L}} + A_{\bar{L}}) + A_{AI}} \quad (MR) \quad (48)$$

$$\frac{K^*(\bar{x})}{K^*} = \frac{\left[\int_{\bar{x}}^{+\infty} f(x;\tilde{\lambda}, k)dx \right] A_{AI} + \alpha A_{\bar{L}}}{\alpha (A_{\underline{L}} + A_{\bar{L}}) + A_{AI}} \quad (OR) \quad (49)$$

Regardless of how labor is reassigned to new tasks, the steady state stock of capital corresponds to

$$K^* = \left(\frac{\alpha}{\rho + \delta} \right)^{1/(1-\alpha)} (A_{\underline{L}} + A_{AI} + A_{\bar{L}}) \quad (50)$$

Expressions (46) to (49) can be replaced in (44) and (45) to obtain the steady state shares of income of each group of agents. The computation yields

$$\frac{\underline{Y}^*}{Y^*} = \frac{A_{\underline{L}} + \left[\int_0^{\underline{x}} (1 + \hat{\zeta})f(x)dx \right] A_{AI}}{A_{\underline{L}} + A_{AI} + A_{\bar{L}}} \quad (MR) \quad (51)$$

$$\frac{\underline{Y}^*}{Y^*} = \frac{A_{\underline{L}} + \left[\int_0^{\underline{x}} f(x;\hat{\lambda}, k)dx \right] A_{AI}}{A_{\underline{L}} + A_{AI} + A_{\bar{L}}} \quad (OR) \quad (52)$$

$$\frac{\bar{Y}^*}{Y^*} = \frac{\left[\int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta}) f(x) dx \right] A_{AI} + A_{\bar{L}}}{A_{\bar{L}} + A_{AI} + A_{\bar{L}}} \quad (MR) \quad (53)$$

$$\frac{\bar{Y}^*}{Y^*} = \frac{\left[\int_{\bar{x}}^{+\infty} f(x; \tilde{\lambda}, k) dx \right] A_{AI} + A_{\bar{L}}}{A_{\bar{L}} + A_{AI} + A_{\bar{L}}} \quad (OR) \quad (54)$$

Steady state income can be derived from the above expressions, and it is not contingent on the labor reinstatement process (note the similarities with the expression of steady state income prior to automation, (13)),

$$Y^* = \left(\frac{\alpha}{\rho + \delta} \right)^{\alpha/(1-\alpha)} (A_{\bar{L}} + A_{AI} + A_{\bar{L}}) \quad (55)$$

From ratios (51) to (54), one may compute the advantage of high-skilled workers over low-skilled workers, i.e., the skill premium. In this setup, the skill premium is not only associated with wages but also with the returns from automation and AI that each group of agents can access,

$$\frac{\bar{Y}^*}{Y^*} = \frac{\left[\int_{\bar{x}}^{+\infty} (1 + \tilde{\zeta}) f(x) dx \right] A_{AI} + A_{\bar{L}}}{A_{\bar{L}} + \left[\int_0^{\bar{x}} (1 + \hat{\zeta}) f(x) dx \right] A_{AI}} \quad (MR) \quad (56)$$

$$\frac{\bar{Y}^*}{Y^*} = \frac{\left[\int_{\bar{x}}^{+\infty} f(x; \tilde{\lambda}, k) dx \right] A_{AI} + A_{\bar{L}}}{A_{\bar{L}} + \left[\int_0^{\bar{x}} f(x; \hat{\lambda}, k) dx \right] A_{AI}} \quad (OR) \quad (57)$$

Appendix 3 illustrates the results by recovering the numerical example.

7 Dynamics: the Expansion of the AI Frontier

Thus far, the analysis has contemplated a one-off round of substitution of labor by AI. In this section, the possibility of a systematic relocation of tasks is raised, following a rule that involves a symbiotic relation between human capital and automation: qualified workers can push the AI frontier outwards, and the new AI frontier shifts part of the labor force in the direction of more qualified jobs, turning this, eventually, into an everlasting expansion process, i.e., on a sustained growth mechanism.

The first step in the dynamic analysis consists in assuming that the value of the parameter associated with the costs of creativity in AI, X , increases with the number of qualified workers (meaning that the creativity costs fall). Qualified workers are those that locate on the right part of the labor productivity distribution once this is broken in two by the intermediate loss of jobs to machines. Analytically, the

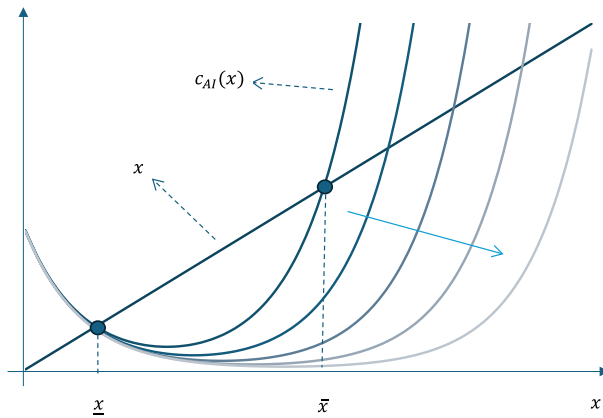


Fig. 4 Effect of consecutive falls in automation and AI costs

following difference equations reflect these dynamics, for both labor reinstatement assumptions,

$$\frac{X_{t+1} - X_t}{X_t} = a \int_{\bar{x}_t}^{+\infty} (1 + \tilde{\zeta}_t) f(x) dx, \quad a > 0, X_0 \text{ given (MR)} \quad (58)$$

$$\frac{X_{t+1} - X_t}{X_t} = a \int_{\bar{x}_t}^{+\infty} f(x; \tilde{\lambda}, k) dx, \quad a > 0, X_0 \text{ given (OR)} \quad (59)$$

The increase in X lowers AI costs, making $\underline{x}$ and $\bar{x}$ change over time. Specifically, as X rises, interval $(\underline{x}, \bar{x})$ expands. The impact of the progressive fall in automation and AI costs is depicted in Fig. 4.¹⁰ As the figure shows, the main effect occurs on the right-end of the distribution, i.e., regarding the costs of AI associated with creative tasks; the impact on the left-end of the productivity distribution is negligible.

For any time period, $A_{L,t}$ and $A_{H,t}$, as already defined, continue to represent the productivity levels associated with low-skilled and high-skilled jobs. Machine productivity, $A_{AI,t}$, is now harder to quantify (and its evolution obeys to a dynamic rule). From (21), the initial value of this term is:

$$A_{AI,0} = \int_{x_0}^{\bar{x}_0} f(x) [x - c_{AI}(x)] dx \quad (60)$$

In the second period, $t = 1$, one must add, to the previous value, the gains from the new activities that machines are able to conquer, i.e.,

¹⁰ The dynamics in Fig. 4 represent, indistinctly, the motion triggered by Eqs. (58) or (59).

$$\begin{aligned}
 A_{AI,1} = & A_{AI,0} + \int_{\underline{x}_1}^{\underline{x}_0} (1 + \hat{\xi}) f(x) [x - c_{AI}(x)] dx \\
 & + \int_{\bar{x}_0}^{\bar{x}_1} (1 + \tilde{\xi}) f(x) [x - c_{AI}(x)] dx \quad (MR)
 \end{aligned}
 \tag{61}$$

$$\begin{aligned}
 A_{AI,1} = & A_{AI,0} + \int_{\underline{x}_1}^{\underline{x}_0} f(x; \hat{\lambda}, k) f(x) [x - c_{AI}(x)] dx \\
 & + \int_{\bar{x}_0}^{\bar{x}_1} f(x; \tilde{\lambda}, k) f(x) [x - c_{AI}(x)] dx \quad (OR)
 \end{aligned}
 \tag{62}$$

Generalizing for period $t + 1$,

$$\begin{aligned}
 A_{AI,t+1} = & \sum_{i=0}^t A_{AI,i} + \int_{\underline{x}_{t+1}}^{\underline{x}_t} (1 + \hat{\xi}) f(x) [x - c_{AI}(x)] dx \\
 & + \int_{\bar{x}_t}^{\bar{x}_{t+1}} (1 + \tilde{\xi}) f(x) [x - c_{AI}(x)] dx \quad (MR)
 \end{aligned}
 \tag{63}$$

$$\begin{aligned}
 A_{AI,t+1} = & \sum_{i=0}^t A_{AI,i} + \int_{\underline{x}_{t+1}}^{\underline{x}_t} f(x; \hat{\lambda}, k) f(x) [x - c_{AI}(x)] dx \\
 & + \int_{\bar{x}_t}^{\bar{x}_{t+1}} f(x; \tilde{\lambda}, k) f(x) [x - c_{AI}(x)] dx \quad (OR)
 \end{aligned}
 \tag{64}$$

The devised dynamic framework is a growth model, for which one can characterize the respective BGP. This characterization is postponed to the next section. The graphical characterization in Appendix 4 prepares the ground for such analysis by identifying patterns of evolution of the main variables, both in the transitional dynamics phase and in the long-term BGP.

8 Balanced Growth Path under the H-AI Symbiosis

The brief numerical example explored in Appendix 4 partially unveils what the long-term balanced growth path looks like. In the BGP we should expect:

1. x^* to be a constant positive value;
2. $\bar{X}^*$ and $\bar{x}^*$ to grow at a same constant rate;
3. A_L^* to be constant;
4. A_{AI}^* and A_L^* to grow at the same constant rate as X^* ;
5. K^* and Y^* to grow at the same constant rate as X^* .

These outcomes are common to the various scenarios under appreciation, regarding capital ownership, the shape of the labor productivity distribution, and the process of labor reassignment.

To confirm that $\underline{x}^*$ is a constant value, note that it is the solution of $\underline{x}^* = \omega e^{-\omega \underline{x}^*} + \omega e^{-\omega(X^* - \underline{x}^*)}$, and that if $\lim_{t \rightarrow +\infty} X_t = +\infty$ then $\lim_{t \rightarrow +\infty} \omega e^{-\omega(X_t - \underline{x}_t)} = 0$. This signifies the equation can be simplified to $\underline{x}^* = \omega e^{-\omega \underline{x}^*}$. Although this last equality cannot be explicitly solved in general form, it is true that for any positive value of ω , it allows for the determination of a unique finite solution for $\underline{x}^*$. The BGP growth rate for X is derivable from (58) and (59). This can be presented for the two assumed prototype productivity distributions. In the case of the exponential distribution (*Exp*),

$$\gamma_X = a(1 - \eta)e^{-\lambda \underline{x}^*} \quad (65)$$

For the Weibull distribution with $k = 2$ (*Wei*),

$$\gamma_X = a(1 - \eta)e^{-(\lambda \underline{x}^*)^2} \quad (66)$$

Growth rates (65) and (66) apply to both labor reinstatement mechanisms assumed throughout the analysis. Observe that the growth rate of X , which is also the growth rate of the economy, is shaped by the four parameters associated with the labor-automation framework, i.e., a, λ, ω, η . Note, in particular, the relevance of parameter η : an increase in the share of workers being relocated to low-skilled tasks (higher η) imposes a fall in the growth rate. Moreover, notice that there is a particular case of zero growth, respecting to the circumstance in which all displaced labor is reinstated at low-skill jobs, i.e., $\eta = 1$.

As stated, growth rate γ_X is also the growth rate of the upper limit $\bar{x}^*$. To verify this result, take the replacement condition for $\bar{x}^*$, i.e., $\bar{x}^* = \omega e^{-\omega \bar{x}^*} + \omega e^{-\omega(X^* - \bar{x}^*)}$. Because now the premise is that $\bar{x}^*$ grows over time, $\lim_{t \rightarrow +\infty} \omega e^{-\omega \bar{x}_t} = 0$ and the only way the relation holds is that if $\bar{x}^*$ grows at the same rate as X^* .

An important result that one can derive for the BGP is the distribution of workers between low-skill and high-skill tasks. These are, respectively,

$$\begin{aligned} \int_0^{\underline{x}^*} f(x)dx + \eta \int_{\underline{x}^*}^{\bar{x}^*} f(x)dx \\ = 1 - (1 - \eta)e^{-\lambda \underline{x}^*} \text{ (Exp)} \\ = 1 - (1 - \eta)e^{-(\lambda \underline{x}^*)^2} \text{ (Wei)} \end{aligned} \quad (67)$$

$$\begin{aligned} (1 - \eta) \int_{\underline{x}^*}^{\bar{x}^*} f(x)dx + \int_{\bar{x}^*}^{+\infty} f(x)dx \\ = (1 - \eta)e^{-\lambda \underline{x}^*} \text{ (Exp)} \\ = (1 - \eta)e^{-(\lambda \underline{x}^*)^2} \text{ (Wei)} \end{aligned} \quad (68)$$

Equations (67) and (68) indicate that the distribution of workers in the BGP depends on the various parameters of the model. Again, particularly relevant is the role of share η , which is essential in splitting workers between regions of low and high productivity.

Next, concentrate on the productivity levels. The analysis is undertaken solely for the exponential distribution, in order to present tractable expressions. The same reasoning applies to the *Wei* case and the same type of qualitative outcome is obtained. Because we know that $\underline{x}^*$ is a constant value and $\bar{x}^*$ grows indefinitely at a constant rate, the following expressions correspond to BGP results,

$$A_L^* = \left(1 + \eta \frac{e^{-\lambda \underline{x}^*}}{1 - e^{-\lambda \underline{x}^*}}\right) \frac{(1 + \lambda \underline{x}^*)e^{-\lambda \underline{x}^*}}{\lambda} \quad (MR) \quad (69)$$

$$A_L^* = \frac{1 - [1 + \lambda \underline{x}^* - \ln(1 - \eta)](1 - \eta)e^{-\lambda \underline{x}^*}}{\lambda \underline{x}^* - \ln(1 - \eta)} \underline{x}^* \quad (OR) \quad (70)$$

$$A_L^* = (1 - \eta) \frac{(1 + \lambda \bar{x}^*)e^{-\lambda \bar{x}^*}}{\lambda} \quad (MR) \quad (71)$$

$$A_L^* = \frac{[1 + \lambda \underline{x}^* - \ln(1 - \eta)](1 - \eta)e^{-\lambda \underline{x}^*}}{\lambda \underline{x}^* - \ln(1 - \eta)} \bar{x}^* \quad (OR) \quad (72)$$

The inspection of productivity levels (69) through (72) indicates that A_L^* is a constant value in both *MR* and *OR* cases; A_L^* grows, in both cases, in the same extent as $\bar{x}^*$. Hence, there is a fundamental difference between low-skilled and high-skilled productivity in the BGP: the former stagnates, while the latter grows at an identical rate to the one pertaining to the expansion of AI tasks.

Concerning the BGP growth rate of A_{AI} , the evaluation of (63) and (64) allows for understanding that this variable continues to grow as long as $\bar{x}^*$ expands, what implies that A_{AI} also grows at rate γ_X .

The following step in the analysis consists in uncovering how growth rates of the productivity terms impact growth rates of aggregate capital and income. To proceed in this direction, it is necessary to revisit the two proposed scenarios regarding capital ownership. With full separation between capitalists and workers, the BGP income level is given by Eq. (34). Because ratio $\frac{\alpha(A_L + A_L) + A_{AI}}{A_L + A_{AI} + A_L}$ is constant in the BGP and, in the same circumstance, term $A_L + A_{AI} + A_L$ grows at rate γ_X , this will also be the growth rate of Y^* . Because of the Cobb-Douglas shape of the production function, also K^* grows at rate γ_X . For the economy of optimal planners, the reasoning is even more straightforward; looking at expressions (50) and (55), one confirms that, also in this case, K^* and Y^* grow at rate γ_X .

Although there is no difference across cases in what concerns the growth rate of aggregate capital and aggregate income, the same is not true when one

Table 2 Synthesis of BGP results (BGP growth rates)

<i>BGP growth rates</i>	Hand-to-mouth consumers + Representative capitalist	Pervasive optimal planning
Productivity of low-skilled workers	0	0
Productivity of high-skilled workers	γ_X	γ_X
Aggregate income	γ_X	γ_X
Income of low-skilled workers	0	γ_X
Income of high-skilled workers	γ_X	γ_X

γ_X is given by equation (65) in the *Exp* case, and by equation (66) in the *Wei* case

evaluates the returns of each class of agents. Again, take first the scenario of separation between workers and the representative capitalist. Recall that, in this case, wages and capital returns are given by expressions (23) and (24). From (24), it is straightforward to realize that the returns of the capitalist should increase as much as income, and therefore, in the BGP, these returns also grow at the economy's growth rate, γ_X .

Wages are split between low-skilled and high-skilled workers through the following terms: $W_L = \frac{(1-\alpha)A_L}{A_L+A_{AI}+A_{\bar{L}}}Y$ and $W_{\bar{L}} = \frac{(1-\alpha)A_{\bar{L}}}{A_L+A_{AI}+A_{\bar{L}}}Y$. Observe that the numerator of the quotient in W_L does not grow in the BGP, while the denominator grows at the same rate as income. Hence, in the BGP, the income of the low-skilled workers (which corresponds to their wage) stagnates. Regarding the wages of the high-skilled, the quotient in the expression does not grow, and therefore $W_{\bar{L}}$ grows at an identical rate as income.

Dividing (24) by (23), one obtains, in this setting, the ratio between capital returns and wages. This is:

$$\frac{rK + \frac{\partial Y}{\partial A_{AI}}A_{AI}}{W} = \frac{\alpha}{1-\alpha} + \frac{1}{1-\alpha} \frac{A_{AI}}{A_L + A_{\bar{L}}} \quad (73)$$

The above expression has two terms. The first is the capital – labor returns ratio in the absence of AI. In this case, the percentage of income destined to each type of agent depends solely on the output-capital elasticity. A positive A_{AI} introduces a new term that makes returns to capital to be higher relatively to the returns on labor. The larger is A_{AI} relative to $A_L + A_{\bar{L}}$, the wider is the gap between $rK + \frac{\partial Y}{\partial A_{AI}}A_{AI}$ and W in favor of $rK + \frac{\partial Y}{\partial A_{AI}}A_{AI}$. In the BGP, A_{AI} and A_L grow at the same rate; hence, the presented ratio tends to a constant value.

Concerning the optimal planners case, there are two groups of agents, low-skilled workers and high-skilled workers, with both groups benefiting from returns to physical capital and automation capital. The returns of these two groups can be obtained from expressions (51) and (53), given aggregate income (55). Examining the expressions, we conclude that both $\underline{Y}^*$ and $\bar{Y}^*$ grow at rate γ_X .

Hence, there is a distinction between the two scenarios; if all agents optimally save, all agents may take advantage from the automation and AI economy. In contrast, if workers have no possibility, capacity, or willingness to save, part of these agents, namely the low-skilled, will be penalized by the progressive automation of production and will fall behind.

This fundamental difference is materialized on the skill premium. Recall that, in the case of separation of capital and labor, $\frac{w_L}{w_H} = \frac{A_L}{A_H}$. This skill premium grows indefinitely because, in the BGP, A_L is constant and A_H exhibits a positive and constant rate of growth. In the setting with agents that simultaneously work and invest, the skill premium is given by expressions (56) and (57). These values are both constant and, therefore, the gap between low-skilled and high-skilled does not widen over time once the BGP position has been reached.

Table 2 summarizes the growth equilibrium results in this section.

9 Conclusion

The model in this paper adds knowledge to the voluminous literature on automation and growth, which was selectively reviewed in Sect. 2. The devised framework builds upon a labor heterogeneity assumption, with workers being assigned to a pre-specified distribution of productivity levels. From the preceding analysis, one may highlight the following ideas:

- (i) *Polarization implies a relocation or reinstatement process.* Workers that no longer can compete with machines in the middle-productivity tasks may migrate in two directions: they can settle in performing jobs that are beneath their capabilities and that, therefore, are also poorly paid. Other workers will adapt to the new economy, acquiring new skills and moving to the right in the productivity distribution. This type of reordering of the labor force will determine how the economy will perform: the larger the share of agents moving to the right along the distribution (i.e., towards high-skilled jobs), the more the economy exhibits potential to grow.
- (ii) *The symbiotic relation between human capital (high-skilled labor) and automation (AI) generates endogenous growth.* Progressively larger availability of human capital lowers the creativity costs associated with the development of robots and algorithms. Simultaneously, better AI leads to a stronger reinstatement of workers in high-skilled activities. This allows to form a virtuous loop that culminates with endogenous growth.
- (iii) *The emergence of labor-saving technologies leads to important redistributive effects.* If there is a complete segregation between workers and capital owners, then workers in the low-skill part of the productivity distribution will receive, in the long-term, a constant wage, and therefore they reap no gain from the automation process. The gains are concentrated on the representative capitalist and on the well-educated workers, whose returns grow at the same BGP growth rate as the economy. On the contrary, if optimal behavior is pervasive,

everyone has the chance to accumulate wealth and access capital returns. Hence, the income of all agents, regardless of the position they occupy in the productivity distribution, will grow as much as the economy and, therefore, everyone benefits from the adoption of new technologies.

One could infer from the undertaken discussion that the solution for the income distribution issues that automation may raise resides on letting and stimulating every agent to formulate optimal consumption-savings plans. Although this is in fact desirable, it is not always possible, for a variety of reasons. First, the implementation of an optimal plan may not be feasible because at every period there is a minimal subsistence consumption level that must be met (in many circumstances, poor people simply cannot save). Second, there are financial literacy issues to consider: even if people can save, they may not have the adequate tools to formulate optimal plans. Third, people may have the means and the knowledge to save but they eventually lack the self-control ability to do so.

In this debate, public intervention emerges as a relevant element at various levels. First, it is important in order to assist workers in shifting to high-quality jobs instead of occupations requiring low skills, whenever automation and AI imply a reconfiguration of the labor market. Second, public policies should be implemented to foster financial literacy and raise the awareness about the benefits of saving. Third, pure redistributive policies are in any case relevant to mitigate the potential negative impact of automation and to make the gains of AI available, in some extent, to all members of society.

Appendix 1—Solution of the Optimal Control Problem

To derive Eq. (6), we write the problem's current-value Hamiltonian function,

$$H(K_t, P_t, C_t) = \ln(C_t) + \frac{1}{1+\rho} P_{t+1} [(r_t - \delta)K_t - C_t] \quad (74)$$

with P_t the shadow price associated with the state variable, i.e., the stock of capital.

The first-order optimality conditions are:

$$\frac{\partial H}{\partial C_t} = 0 \Rightarrow C_t^{-1} = \frac{1}{1+\rho} P_{t+1} \quad (75)$$

$$\begin{aligned} \frac{1}{1+\rho} P_{t+1} - P_t &= -\frac{\partial H}{\partial K_t} \Rightarrow \frac{1}{1+\rho} P_{t+1} \left[1 - \delta + \frac{dr_t}{dK_t} K_t + r_t \right] = P_t \\ &\Rightarrow \frac{1}{1+\rho} P_{t+1} \left[1 - \delta + \alpha^2 \left(\frac{K_t}{A_L} \right)^{-(1-\alpha)} \right] = P_t \end{aligned} \quad (76)$$

Replacing (75) into (76), one arrives to the expression of the dynamic rule for consumption in Eq. (6).

Appendix 2—Income Distribution with Hand-to-Mouth Consumers: a Numerical Example

Recover the parameter values of the numerical illustration in Sect. 5 and take, additionally, the following values associated with the intertemporal optimization problem: $\alpha = 1/3$, $\rho = 1/27$, $\delta = 1/81$; consider, as well, that three quarters of the displaced labor fall back to low-skilled jobs, i.e., $\eta = 0.75$. With this parameterization, the steady state income level with no automation in (7) is $Y^* = 1.5$ (*Exp*) or $Y^* = 1.3293$ (*Wei*). With automation, four different outcomes are attained, considering the two distributions and the two labor relocation processes; in every case, these values are larger than those in the no automation scenario: $Y^* = 3.2227$ (*Exp/MR*), $Y^* = 3.3279$ (*Exp/OR*), $Y^* = 2.0686$ (*Wei/MR*), $Y^* = 2.9680$ (*Wei/OR*).

In the no-automation environment, aggregate wages corresponded to two thirds of income, i.e., $W^* = 1$ (*Exp*), $W^* = 0.8862$ (*Wei*); the capital return was one third of income, $(rK)^* = 0.5$ (*Exp*), $(rK)^* = 0.4431$ (*Wei*). With automation, under the assumed example, wages are given by (23), $W^* = 1.2370$ (*Exp/MR*), $W^* = 1.3157$ (*Exp/OR*), $W^* = 1.0062$ (*Wei/MR*), $W^* = 1.6293$ (*Wei/OR*). Automation leads, in any of the settings, to a fall on the weight of wages in total income; from the 66.(6)% prior to automation, such weight falls to the following percentages: 38.38% (*Exp/MR*), 39.54% (*Exp/OR*), 48.64% (*Wei/MR*), 54.89% (*Wei/OR*). Evidently, the outcome is the opposite in what respects capital gains; from the initial 33.(3)%, total capital returns evolve to the following values and percentages, for each of the cases under appreciation: $\left(rK + \frac{\partial Y}{\partial A_{AI}} A_{AI}\right)^* = 1.9857$ (61.62% - *Exp/MR*); $\left(rK + \frac{\partial Y}{\partial A_{AI}} A_{AI}\right)^* = 2.0122$ (60.46% - *Exp/OR*); $\left(rK + \frac{\partial Y}{\partial A_{AI}} A_{AI}\right)^* = 1.0624$ (51.36% - *Wei/MR*); $\left(rK + \frac{\partial Y}{\partial A_{AI}} A_{AI}\right)^* = 1.3388$ (45.11% - *Wei/OR*). One realizes that, independently of the assumed productivity distribution and labor relocation process, both wages and capitalist's return increase with automation; however, these increases are uneven, with capital gains rising more than wages.

Among workers, there is also a substantial change in the structure of earnings: because some workers are pushed into low productivity jobs while others are pushed in the opposite direction, income inequality increases, and therefore, depending on the capacity to adapt to the new economy, there will be losers and winners among the labor class. Because the labor force is now split in two separate groups, one can compute a skill premium, as the ratio between the wages of high-skilled workers and the wages of low-skilled workers. Given (23), this is such that $\frac{w_H}{w_L} = \frac{A_H}{A_L}$. Proceeding with the computation, the outcome is: $\frac{w_H}{w_L} = 3.0126$ (*Exp/MR*), $\frac{w_H}{w_L} = 4.2679$ (*Exp/OR*), $\frac{w_H}{w_L} = 1.4498$ (*Wei/MR*), $\frac{w_H}{w_L} = 6.7960$ (*Wei/OR*). Results on the skill premium differ substantially across cases, namely in what concerns the Weibull distribution. This is the direct outcome of the mass of workers shifting in one or the other direction regarding the possible acquisition of new skills. As shown in Fig. 2, in the *OR* scenario, the number of workers falling to low-skilled jobs is larger than in the *MR* case, what justifies the larger skill premium in the *OR* case.

Appendix 3—Income Distribution with Optimal Planners: A Numerical Example

Given the same parameter values as in the hand-to-mouth scenario (Appendix 2), the steady state income levels under (55) are, in the full optimization case: $Y^* = 4.1056$ (*Exp/MR*), $Y^* = 4.2797$ (*Exp/OR*), $Y^* = 2.8865$ (*Wei/MR*), $Y^* = 4.4193$ (*Wei/OR*). As expected, these values are, for every scenario, larger than the ones computed in the case of hand-to-mouth workers. Next, take quotients (51) to (54); the corresponding values, for each of the assumed alternatives, are displayed in Table 3. The shape of the labor productivity distribution and the kind of labor reassignment process matter for the extent in which income is distributed across classes of agents.

Formulas (56) and (57) allow for the computation of the skill premium. Results are: $\frac{\bar{Y}^*}{Y^*} = 1.0491$ (*Exp/MR*), $\frac{\bar{Y}^*}{Y^*} = 1.2569$ (*Exp/OR*), $\frac{\bar{Y}^*}{Y^*} = 0.9152$ (*Wei/MR*), $\frac{\bar{Y}^*}{Y^*} = 3.2458$ (*Wei/OR*). Comparing these results with the wage skill-premium of the scenario with separation between workers and capitalist (in Appendix 2), one realizes that results are, now, all lower than before. This is a non surprising result, in the sense that if workers access capital returns in an optimal way, this helps in compensating the wage gap between groups, leading to a more balanced distribution of returns. If everyone makes optimal consumption-savings plans, the gains from AI become more evenly distributed.

Appendix 4—Graphical Inspection of Growth Dynamics

For the parameter values used in the numerical examples throughout the paper, one can simulate the growth model in Sect. 7. The simulation delivers the time trajectories displayed in Fig. 5. The graphics in the figure were constructed taking as reference case *Exp/MR*; no significant qualitative differences exist if the graphics are drawn for any of the other assumed possibilities.

The figure contains three panels. In the first panel, the time trajectories of the growth rate of X and of the threshold values $\underline{x}$ and $\bar{x}$ are displayed. The visual analysis suggests that $\underline{x}$ tends to a constant long-term value, while X and $\bar{x}$ will grow at the same positive constant rate, once these variables reach the BGP. The second panel exhibits the growth rates of each of the productivity components. In the long-run, A_L does not grow, and A_{AI} and A_L grow at the same positive constant rate; this rate is identical to the growth rate of X . Finally, the third panel shows the growth rate

Table 3 Income ratios in the full optimization case

	$\frac{Y^*}{Y}$	$\frac{\bar{Y}^*}{Y}$
<i>Exp / MR</i>	0.4880	0.5120
<i>Exp / OR</i>	0.4431	0.5569
<i>Wei / MR</i>	0.5221	0.4779
<i>Wei / OR</i>	0.2355	0.7645

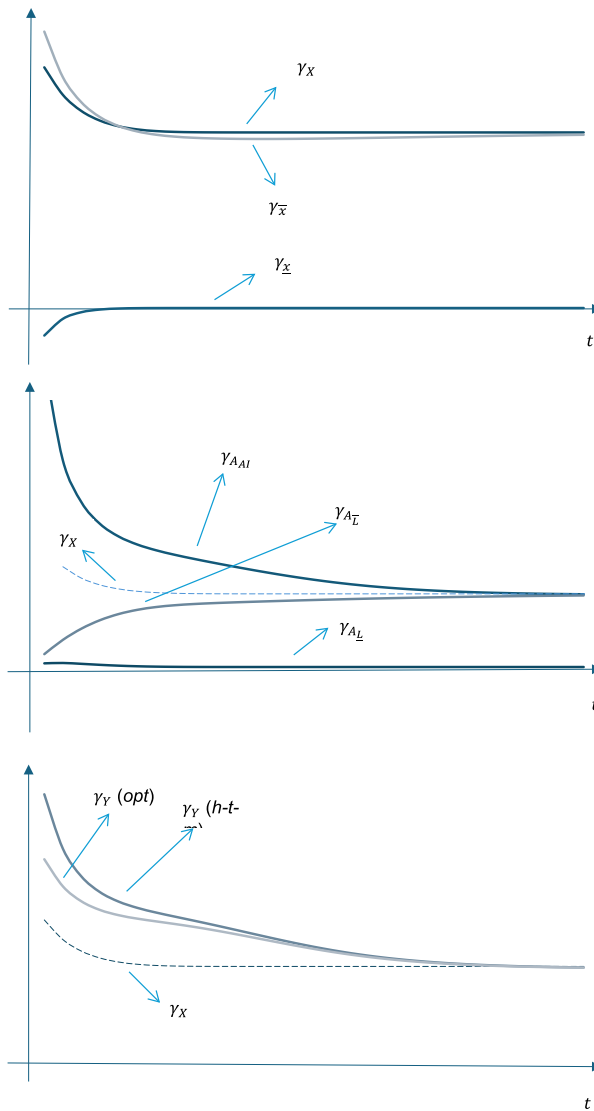


Fig. 5 Transitional dynamics

of aggregate income, which, in the BGP, is also positive, constant, and identical to the growth rate of X .¹¹

¹¹ The trajectory of the economy's growth rate is presented for both considered cases regarding capital ownership.

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